Four follow-ups from the 102-asset study

Rolling strategy choice, a correlation-capped portfolio, regime switching, and daily bars.

Research prototype. Not investment advice. October 2026.

1. What was tested

The 102-asset report left four items open: rolling strategy re-selection, a multi-asset portfolio with a correlation constraint, regime-conditional switching inside one asset, and daily bars for intraweek signals. This note runs those four on the data already computed.

- SPY sleeves are rebuilt from the saved daily, weekly, and mixed predictions. The 2013-onward Sharpe matches the published file to 0.001.
- A choice uses only bars before the window it trades. Thresholds and scales are the frozen values from each saved summary.
- TASI uses the saved period scores. The strategy for a period is the winner of the previous period.
- The portfolio is a buy-and-hold book of the 102 names. Per-asset strategy paths were not stored, so the optimizer cannot apply those sleeves.
- Oracle rows pick the sleeve with the best Sharpe inside the window they score. They are an upper bound.

Cost is 2 bps of the position change, the same convention as the earlier studies. Sharpe uses sqrt(252) on daily bars and sqrt(52) on weekly bars. CAGR uses calendar time.

2. Rolling strategy re-selection

Every quarter on daily bars, and every 13 weeks on weekly bars, the book takes the sleeve with the highest Sharpe over the trailing year. The pre-2013 winner is the frozen rule.

Daily

Rule	Sharpe	CAGR %	Max DD %
Binary (t=0)	0.915	13.16	-19.5
Binary (t=opt)	1.041	15.99	-24.4
Scaled	0.694	6.53	-15.5
Adaptive	0.712	8.62	-22.6
Buy and hold	0.901	14.61	-33.7
Frozen (Binary (t=opt))	1.041	15.99	-24.4
Rolling	0.954	13.07	-17.6
Oracle	1.405	18.57	-17.6

The rolling book switched 18 times. Share of bars: Binary (t=opt) 60%, Adaptive 14%, Scaled 14%, Binary (t=0) 11%. The frozen pre-2013 winner was Binary (t=opt).

Weekly

Rule	Sharpe	CAGR %	Max DD %
Binary (t=0)	0.911	13.64	-31.8
Binary (t=opt)	0.202	1.38	-22.6
Scaled	0.549	5.11	-28.1
Adaptive	0.909	12.60	-31.8
Buy and hold	0.942	14.62	-31.8
Frozen (Binary (t=opt))	0.202	1.38	-22.6
Rolling	0.695	8.77	-28.1
Oracle	1.338	16.49	-22.6

The rolling book switched 14 times. Share of bars: Adaptive 42%, Binary (t=0) 31%, Scaled 14%, Binary (t=opt) 13%. The frozen pre-2013 winner was Binary (t=opt).

Mixed

Rule	Sharpe	CAGR %	Max DD %
Binary (t=0)	0.763	11.29	-33.5
Binary (t=opt)	0.172	1.06	-22.6
Scaled	0.475	5.09	-33.1
Adaptive	0.693	8.82	-38.3
Buy and hold	0.901	14.61	-33.7
Frozen (Binary (t=opt))	0.172	1.06	-22.6
Rolling	0.510	6.03	-41.6
Oracle	1.399	17.74	-21.8

The rolling book switched 25 times. Share of bars: Adaptive 34%, Binary (t=0) 31%, Binary (t=opt) 22%, Scaled 13%. The frozen pre-2013 winner was Binary (t=opt).

On daily SPY, rolling Sharpe is 0.954 against 0.915 for a fixed zero threshold and 1.041 for the frozen optimal threshold. The oracle, which sees each upcoming quarter, reaches 1.405. Adapting the sleeve on trailing Sharpe does not beat the best fixed rule on this path.

TASI, previous period picks the next one

155 names have at least two periods. The 283 rows below are the later periods only. The choice is locked before the period starts.

Rule	N	Med Sh	Med CAGR	Beat BnH	Gap vs t=0
Always t=0	283	0.412	6.48	185/283	+0.000
Previous Sharpe winner	283	0.398	4.20	178/283	+0.000
Previous winner, in market >= 20%	283	0.420	4.88	178/283	+0.000
Frozen first-period winner	283	0.379	4.41	176/283	+0.000
Oracle	283	0.735	9.94	248/283	+0.259

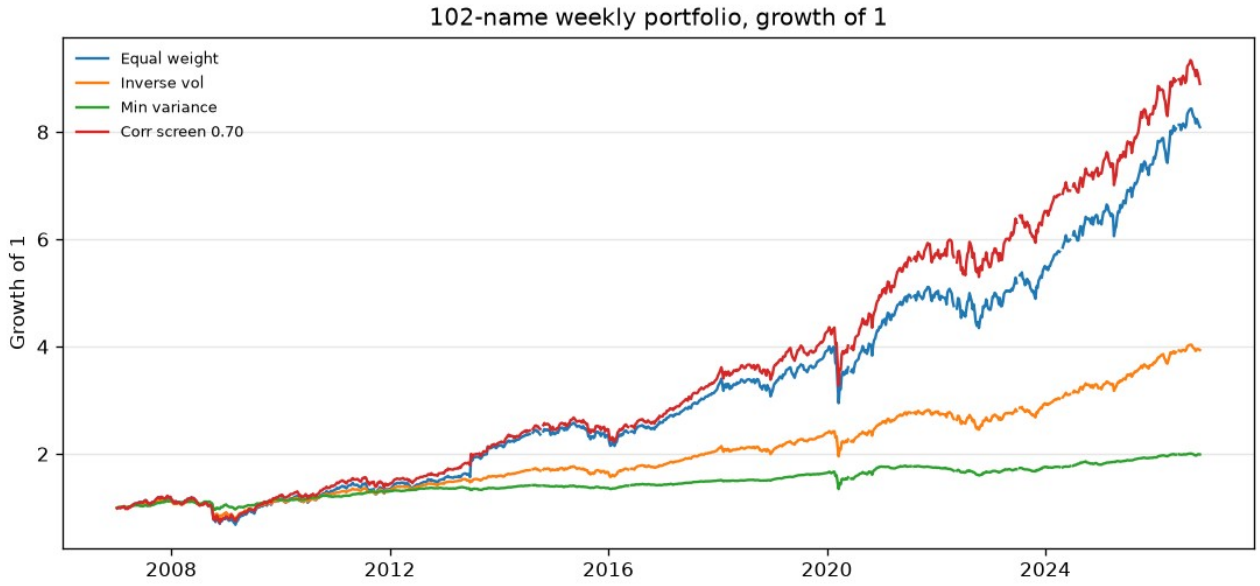
Buy-and-hold median Sharpe on the same periods is 0.286, median CAGR 3.69%. The previous-period winner beats the fixed zero threshold in 85 of 283 periods. The median gap is +0.000. Requiring that winner to have been in the market at least 20% of sessions beats the zero threshold in 86 of 283 periods, median gap +0.000.

3. Portfolio with a correlation constraint

Weekly closes for 102 of the 102 names. Once each January, weights are fit on the prior 156 weeks and held for the next year. 20 rebalances, 2007-01-05 through 2026-01-02. Minimum variance is long-only, 50% shrunk toward its diagonal, with an 8% name cap. The correlation screen starts from every eligible name, drops the higher-volatility name of any pair whose sample correlation is above 0.70, and equal-weights whoever remains. Minimum variance is a different book: it already concentrates in bonds and currencies.

Book	Sharpe	CAGR %	Max DD %	Pair corr	Names	Turnover
Equal weight	0.762	11.13	-43.4	0.30	97	0.11
Inverse vol	0.746	7.16	-31.9	0.32	78	0.17
Min variance	0.606	3.53	-19.4	0.14	21	0.38
Corr screen 0.70	0.864	11.66	-41.2	0.19	46	0.43

Average pairwise correlation is the sample correlation of the names held, on the same past window that set the weights. The January 2026 book put this much weight by class. Min variance: Bond ETF 48%, Currency 41%, Commodity 5%, TASI Stock 3%, US Stock 3%, Intl Stock 1%. Correlation screen: US Stock 27%, TASI Stock 20%, Commodity 12%, Intl Stock 10%, Currency 8%, Intl ETF 7%.



4. Regime-conditional switching

The saved HMM state is known at the close. The return it is allowed to earn is the next bar. Every quarter the map from bear, sideways, and bull to a sleeve is refit on earlier bars only. A state with fewer than 40 past daily bars, or 20 weekly bars, stays on Binary (t=0).

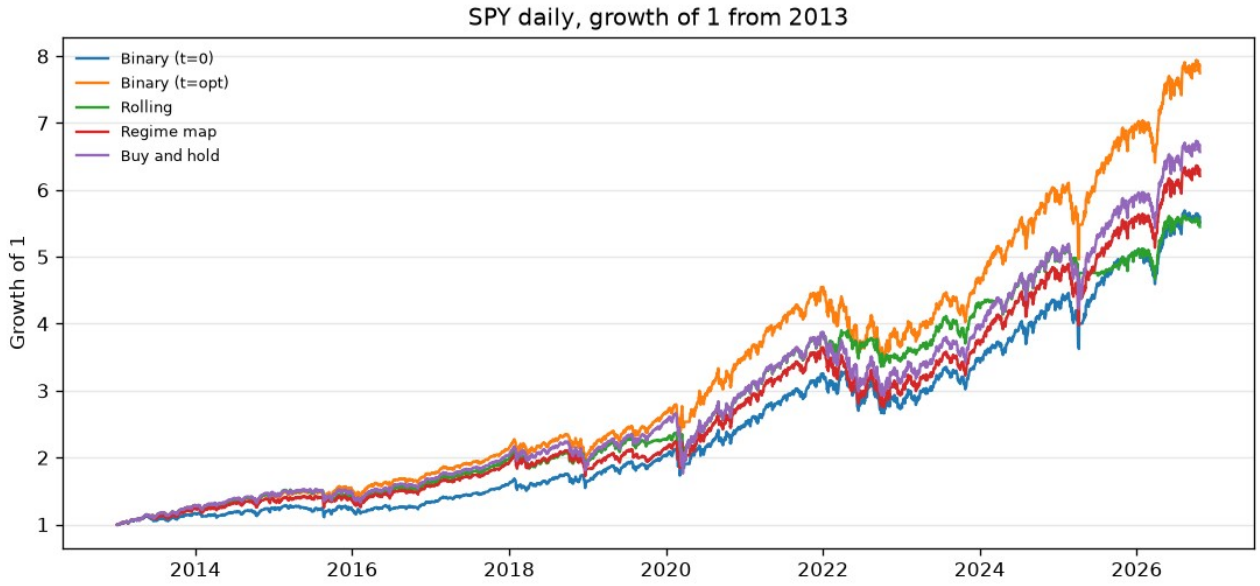
Mode	Regime Sh	CAGR %	Max DD %	t=0 Sh	Rolling Sh	Maps differ
daily	0.948	14.15	-24.4	0.915	0.954	16/56
weekly	0.784	10.37	-31.9	0.911	0.695	56/56
mixed	0.460	5.62	-38.4	0.763	0.510	44/56

Maps differ counts how many refits assigned different sleeves to different states. The last daily map was: Bear -> Binary (t=opt), Sideways -> Binary (t=opt), Bull -> Binary (t=opt).

Where each sleeve sat inside the evaluation window

These Sharpes split the 2013-onward daily path by the HMM state that day. They describe the window. They were not used to place a trade.

Regime	N	t=0	t=opt	Scaled	Adaptive
Bear	1050	1.14	1.27	1.03	0.77
Sideways	892	0.68	0.68	0.18	0.21
Bull	1531	0.95	1.19	0.80	1.10



5. Daily bars and the intraweek path

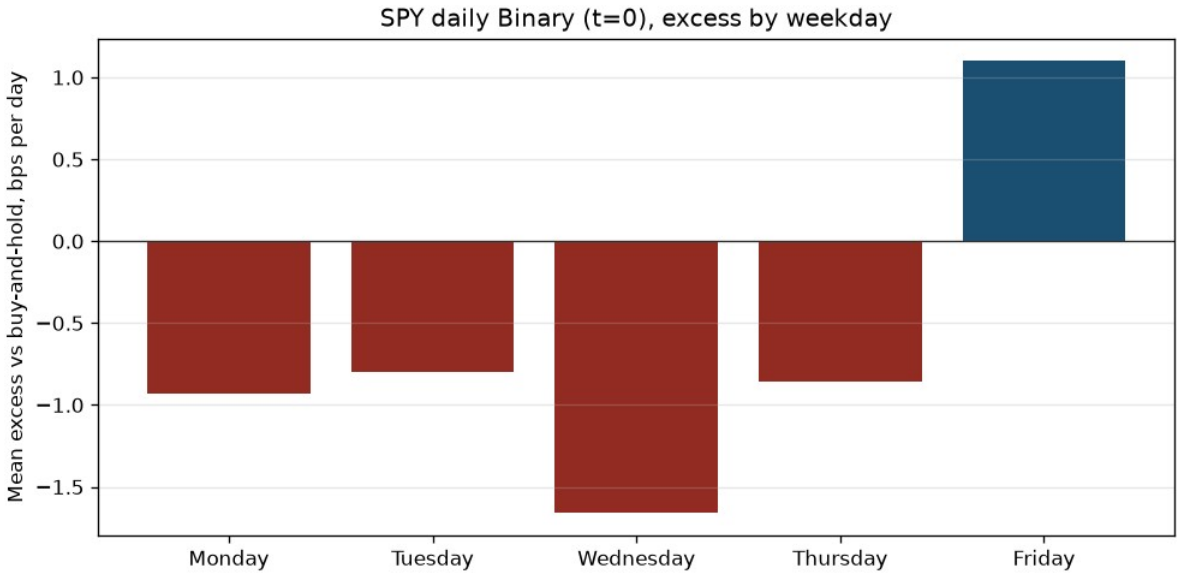
The 102-asset run already scored this. Best daily Sharpe beat best weekly Sharpe on 63 of 102 names, median paired gap +0.066. Binary (t=0) daily beat weekly on 58 names. Median Sharpe was daily 0.556, weekly 0.532, mixed 0.577, buy-and-hold 0.450. Mixed beat the daily book on a paired basis by -0.006, and on 49 names for the zero threshold.

Class	N	Daily>Wk	Daily	Weekly	Mixed
Bond ETF	9	4/9	0.45	0.47	0.39
Broad ETF	5	2/5	0.88	0.86	0.77
Commodity	9	7/9	0.27	0.28	0.29
Currency	8	3/8	0.20	0.15	0.10
Intl ETF	13	9/13	0.51	0.37	0.52
Intl Stock	8	5/8	0.57	0.51	0.59
REIT	4	1/4	0.34	0.51	0.42
Sector ETF	9	5/9	0.81	0.76	0.74
TASI Stock	12	10/12	0.80	0.64	0.79
Thematic ETF	5	5/5	0.47	0.13	0.41
US Stock	20	12/20	0.82	0.76	0.81

SPY excess by weekday

Daily Binary (t=0) minus buy-and-hold, averaged inside each weekday, from 2013 onward. The sum column is the total excess return, in decimal, added on that weekday.

Weekday	N	BnH bps	t=0 excess bps	t=0 sum	t=opt excess bps
Monday	651	9.7	-0.93	-0.060	-1.08
Tuesday	716	7.8	-0.80	-0.057	1.03
Wednesday	712	2.0	-1.66	-0.118	0.68
Thursday	699	5.7	-0.86	-0.060	0.42
Friday	695	5.0	1.10	0.076	0.78



Daily book compounded to weeks, against the weekly book

Each daily sleeve is compounded to a Friday week and aligned with the native weekly sleeve. Buy-and-hold agrees across the two calendars with correlation 0.992.

Sleeve	Weeks	Daily->week Sh	Weekly Sh	Corr	Daily wins
Binary (t=0)	720	0.954	0.912	0.79	265/720
Binary (t=opt)	720	1.091	0.203	0.25	431/720
Adaptive	720	0.705	0.910	0.65	270/720

6. What the four tests say

- On daily SPY, trailing-Sharpe switching finishes at 0.954. A fixed zero threshold is 0.915. The frozen optimal threshold is 1.041. The oracle upper bound is 1.405. On weekly bars the switcher is 0.695 against 0.911 for the zero threshold and 0.942 for buy-and-hold. On the mixed frame it is 0.510 against 0.763. The pre-2013 Sharpe winner on the weekly and mixed frames is the optimal-threshold sleeve, and that sleeve sits in cash.

- On TASI, carrying the previous period's Sharpe winner beats a fixed zero threshold on 85 of 283 later periods. Median Sharpe is 0.398 against 0.412, median gap +0.000, median CAGR 4.20% against 6.48%. The in-market guard is 0.420. The oracle median gap versus the zero threshold is +0.259. The room is real. The previous winner does not collect it.

- Equal weight Sharpe 0.762, CAGR 11.13%, max drawdown -43.4%, average pairwise correlation 0.296. Minimum variance Sharpe 0.606, CAGR 3.53%, max drawdown -19.4%, pairwise correlation 0.14. The 0.70 screen Sharpe 0.864, CAGR 11.66%, max drawdown -41.2%, pairwise correlation 0.194, median names 46.

- Regime switching on daily SPY has Sharpe 0.948. It holds the optimal-threshold sleeve on 96% of bars. 16 of 56 maps used more than one sleeve. Inside the evaluation window that sleeve leads in the bear, sideways, and bull states, so the map has little to switch. Weekly regime switching is 0.784 against 0.911 for the zero threshold.

- Across the 102 names, best daily Sharpe beats best weekly Sharpe on 63 names, median paired gap +0.066. Mixed features change the paired gap by -0.006. On SPY, compounding the daily zero-threshold sleeve to the same Friday-to-Friday week as the weekly model gives Sharpe 0.954 against the native weekly sleeve 0.912, correlation 0.79. The daily optimal-threshold sleeve, compounded the same way, is 1.091 against the native weekly optimal-threshold sleeve 0.203.

These are research results on saved paths and weekly prices. The 2 bp cost is the study convention. The portfolio is not the ML sleeve, because those paths were not saved name by name. One SPY path is one asset.